Siddhi Sanjay More

Robotics | Electronics | Controls and Prototyping

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Summary

MS Robotics candidate at Northeastern (GPA 3.67 · Global Student Award) with demonstrated experience across fabrication, electronics, and embedded systems; applied on real hardware at the EXP Makerspace. Equally comfortable designing a circuit in KiCad, debugging a print failure on the Ultimaker, or mentoring a student through their first prototype. Brings the technical range and teaching instinct a Makerspace Trainer role is built around.

Technical Skills

Fabrication: 3D printers (UltiMaker), Cura, SLA Resin, Laser Cutting, Foam Cutting, Woodworking, CNC Milling (G-code and M-code), Textile Machines, Bambu Studio, SolidWorks, Fusion 360, ANSYS, AutoCAD, CATIA, NX 11, Plastic Injection Molding, Dynamic Simulations

Electronics and Robotics: KiCad, Fritzing, PCB Layout, Arduino, ESP32 (WiFi/Bluetooth), CAN-FD, MPU6050, FSR, 8051, Microcontrollers, Multimeters, Oscilloscopes, Arduino IDE, LTspice, PCB Milling, Soldering (SMD & through-hole), OpenSim, Robo-Analyzer, Circuit Design, Hardware Debugging

Software: Python, C, MATLAB, Linux, Git, ROS/ROS2

Projects

Smart Ankle Prosthetics, *Northeastern University* **Jan 2026 - Apr 2026**

- Fabricated the ankle foot structure from designed CAD files, iteratively **resolved print failures** including warping, layer delamination, and dimensional inaccuracy across multiple **FDM iterations on the Ultimaker using Cura**; used **SLA resin printing** for precision components requiring tighter tolerances
- **Designed FSR voltage divider schematic in KiCad**, laid out breadboard in **Fritzing,** and **soldered ESP32, MPU6050, and FSR** components onto a **PCB board** for data collection; **integrated Moteus C1 brushless motor** controller via **CAN-FD USB** adapter to PC using **Python** and used **multimeter** to run diagnostics and **debug voltage divider issues** (10kΩ corrected to 470Ω)
- Programmed **ESP32-S2 (WiFi/Bluetooth**) in **Arduino IDE** for **200Hz real-time sensor fusion** and CAN-FD motor control; **logged sensor data via Python** over Serial/USB

Semi-Autonomous Aquatic Robot, *Smart India Hackathon* **Jul 2023 - Oct 2023**

- Designed the complete **multi-microcontroller architecture** and drew the full circuit schematic in **EasyEDA** ESP32 as master node controlling **4 Arduino Nano slaves over I2C**, each dedicated to propulsion, drum/crusher, servo, and ultrasonic obstacle avoidance; specified and integrated L298N motor drivers, rotary encoders, and buck converters
- Built and trained a **CNN (TensorFlow 2.0, Keras)** on 991 images for hyacinth vs. clear water classification with **87.5% test accuracy**; ran **SolidWorks FEA (von Mises stress analysis)** on the crusher drum mechanism to validate structural integrity under load

Experience

Robotics Intern, *IIT Bombay & 1Stop.ai*, *India* **Feb 2022 - May 2022**

- Developed **forward and inverse kinematic** solvers for a **4-DOF SCARA robot in MATLAB**; applied **DH parameter analysis** for full workspace characterization and validated motion configurations in **Robo-Analyzer**.
- Independently designed the complete **SCARA robot assembly in SolidWorks:** modeled arm links, base, end effector, and board with full part documentation and multi-view engineering renders; validated structural integrity under load in **ANSYS Workbench**

Team Vaayushastra, *Fr. CRCE, India* **Aug 2021 - Nov 2021**

- Served as Materials In-charge (Junior) for SAE Aero Design West Competition- selected and sourced materials for wing construction, **operated laser cutters and woodworking equipment**, and **performed foam cutting for wing profiles**; designed landing gear and wings in **SolidWorks and ran FEA simulations** to validate structural integrity under competition deadlines

Education

MS in Robotics, *Northeastern University* **May 2027**

Concentration in Mechanical Engineering, GPA: 3.67, Global Student Award, Relevant Courses: Robot Sensing & Navigation, Robotics Mechanics & Control, Assistive Robotics, Wearable Robotics, Control Systems Engineering, Foundations of AI

BE in Mechanical Engineering, *Mumbai University* **Aug 2020 - Jul 2024**

Honor in Robotics, Relevant Courses: Finite Element Analysis, Automation and AI, CAD/CAM, Supply Chain Management, IoT

Mentorship and Engagement

ASME Student Representative, *CRCE Student Chapter, India* **March 2022 - Aug 2022**

Organized and mentored at Mech-a-thon; led technical workshops, developed structured guides for complex topics, and maintained active engagement between teaching faculty and students